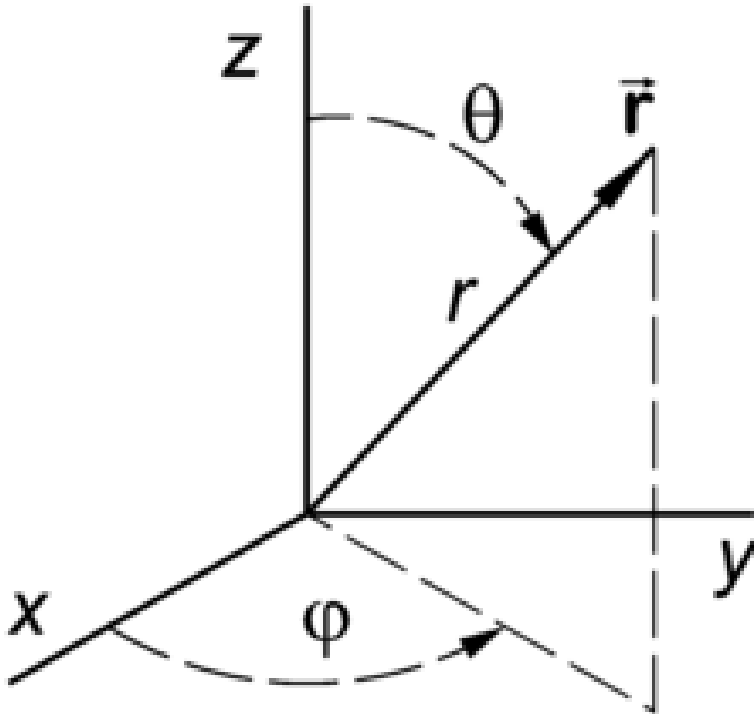


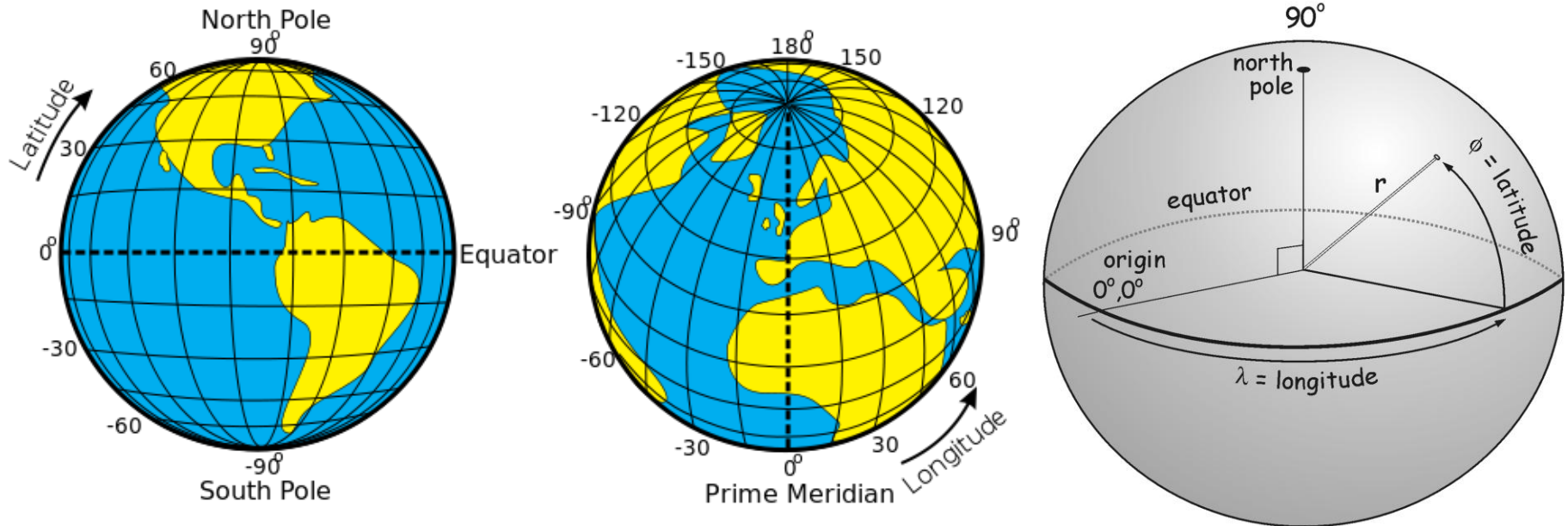
Time, Planning Observations, and Coordinate Systems

Spherical Coordinate Systems

Astronomers use spherical coordinate systems, but many different versions.



1. Longitude + Latitude on Earth

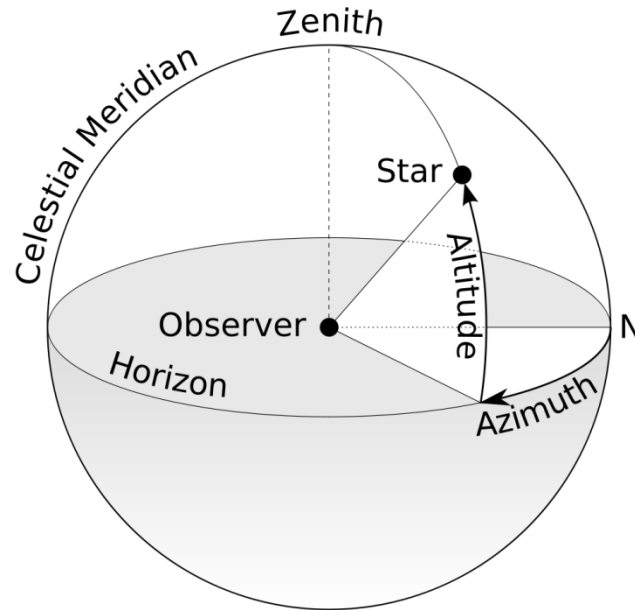


For this system:

- 1) Longitude (longitude)
- 2) Latitude (latitude)
- 3) The Prime Meridian (0 longitude) (Greenwich, England) (increasing CCW from north pole)
- 4) The Equator (0 latitude)
- 5) Origin is the center of the Earth

Poles are north and south pole

2. Horizontal or Alt/Az or El/Az



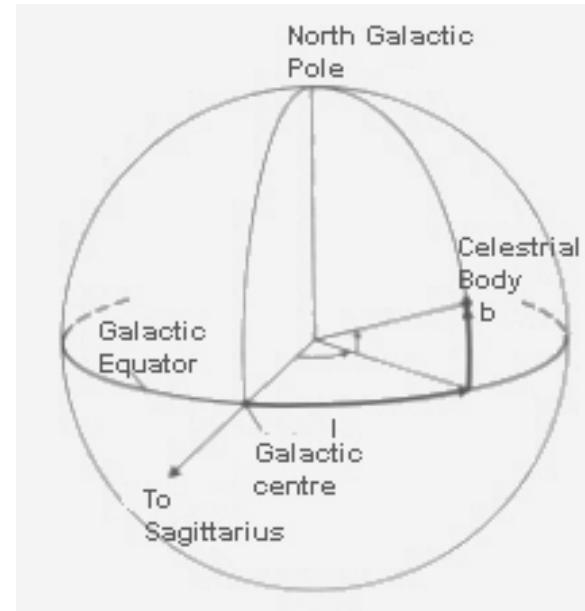
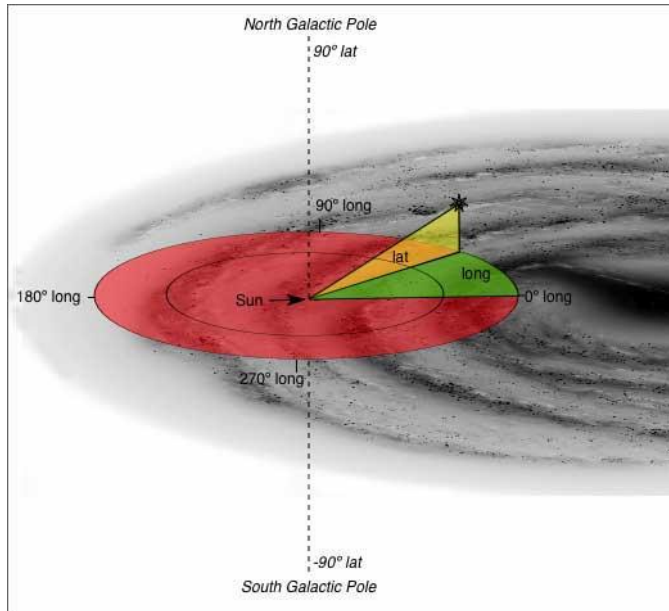
For this system:

- 1) Azimuth (A) (longitude)
- 2) Altitude (a) (latitude)
- 3) North (0 longitude) (increasing CW from zenith)
- 4) The horizon (0 latitude)
- 5) Origin is the observer

Poles are zenith and nadir

Zenith distance is $90 - \text{altitude}$

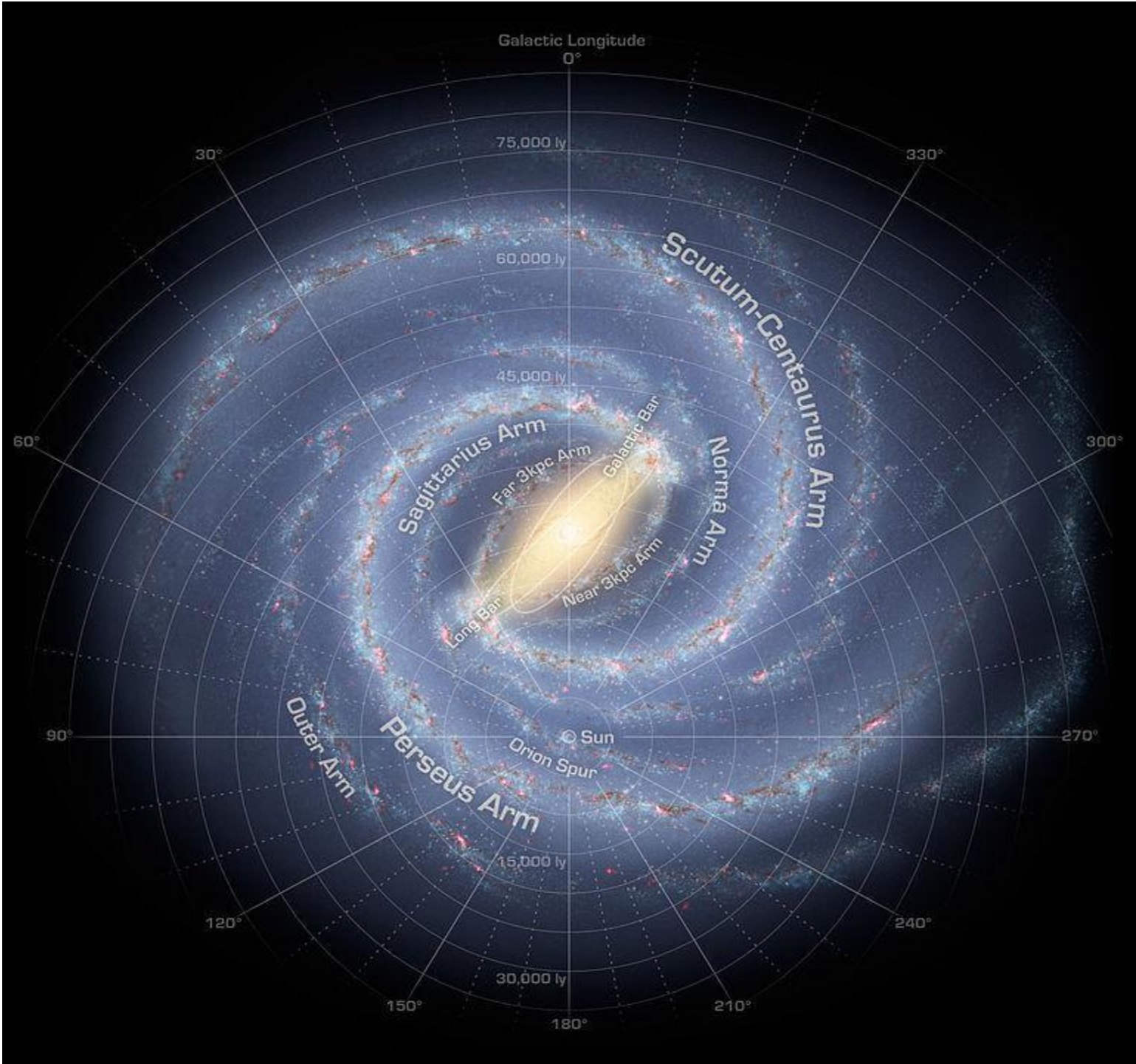
3. Galactic

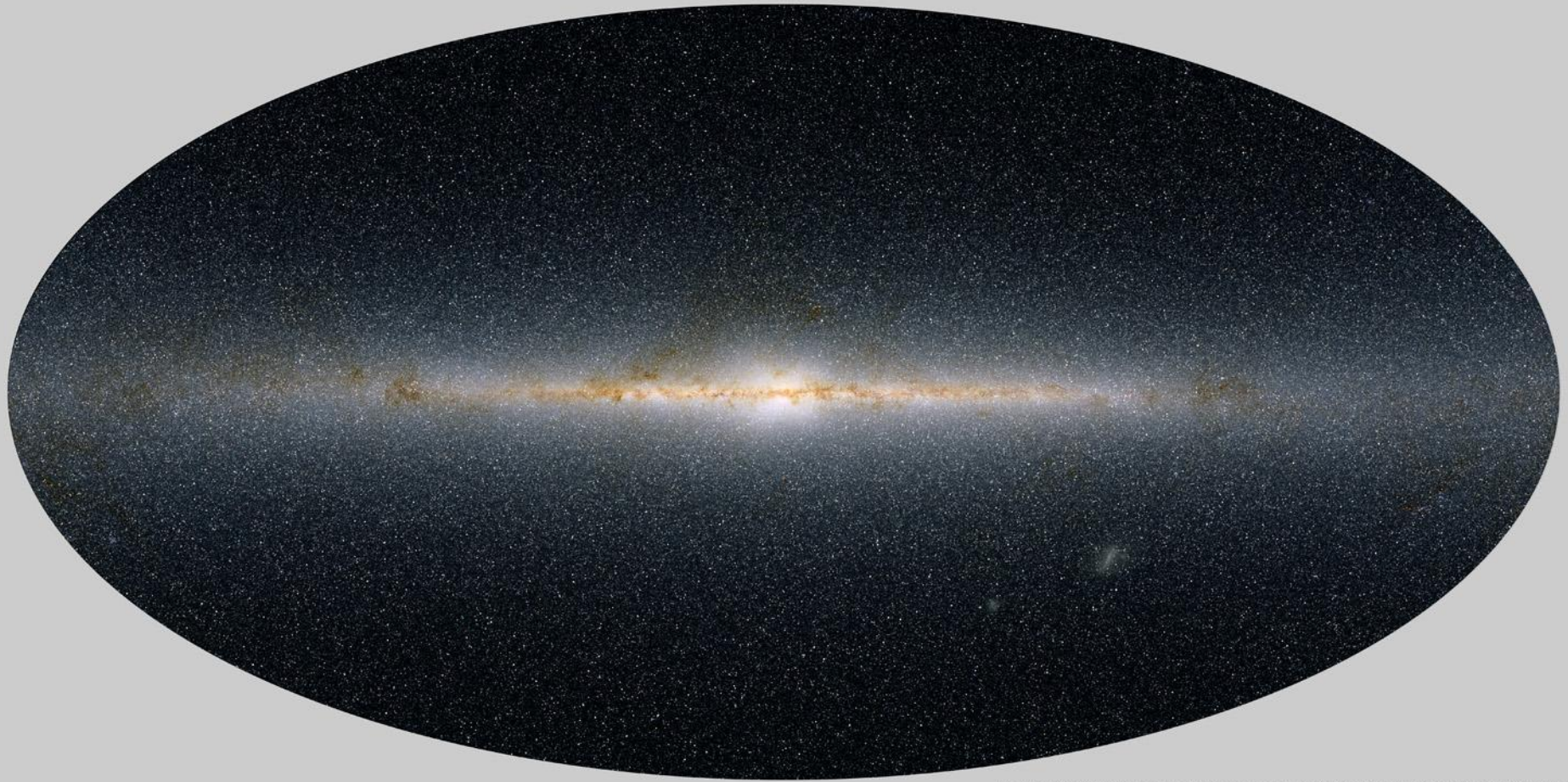


For this system:

- 1) Galactic longitude (l) (longitude) (increases CW from north Galactic pole)
- 2) Galactic latitude (b) (latitude)
- 3) The Galactic center (0 longitude)
- 4) The Galactic plane (0 latitude)
- 5) Center of the Sun

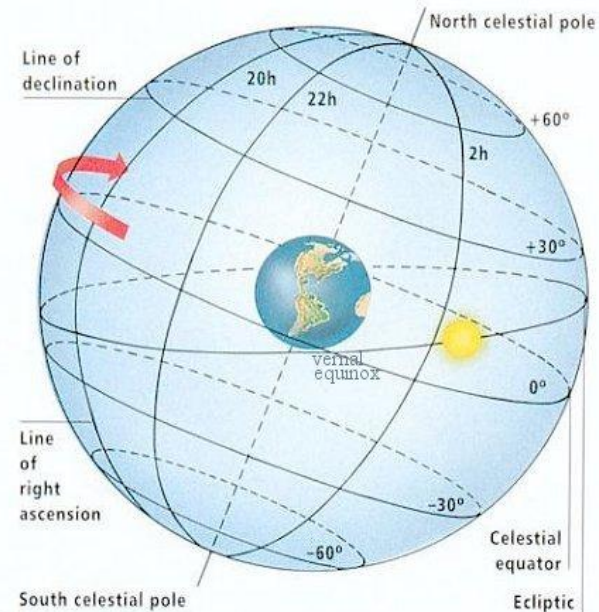
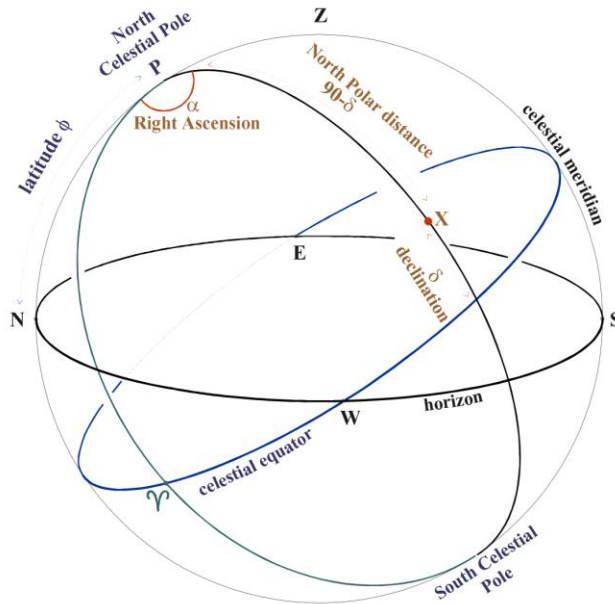
Poles are north and south Galactic poles





Two Micron All Sky Survey Image Mosaic: Infrared Processing and Analysis Center/Caltech & University of Massachusetts

4. Equatorial

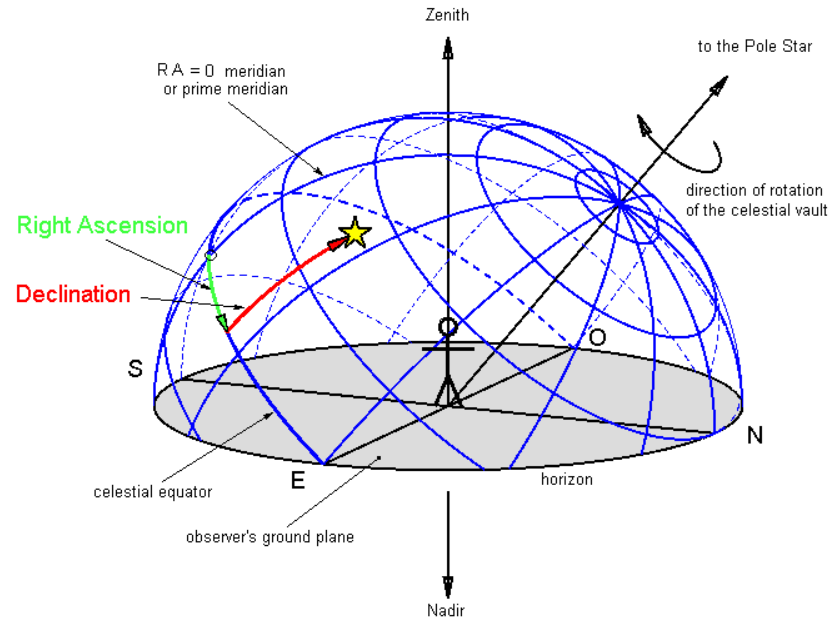


For this system:

- 1) Right Ascension (longitude) (RA, α) (increases CCW from NCP)
- 2) Declination (latitude) (Dec., δ)
- 3) The Vernal Equinox (γ) defines the “prime meridian” (0 longitude)
- 4) The Celestial Equator (0 latitude)
- 5) Center of the Earth OR Center of the Sun

Poles are north (+90 deg. Dec.) and south (-90 deg. Dec.) celestial poles (NCP, SCP)

Declination



The declination measures angular distance from the NCP or SCP

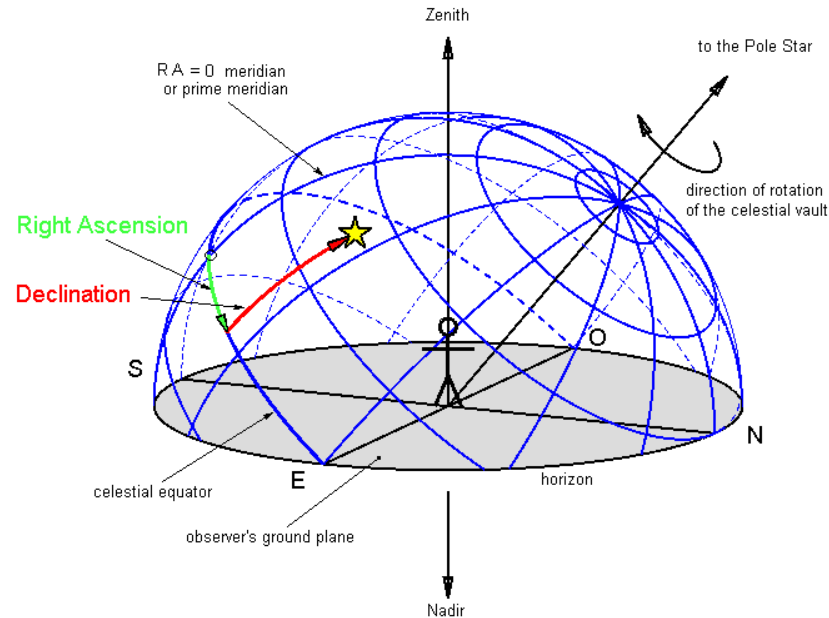
For northern hemisphere:

- Sources with $\text{dec} > 90 - \text{lat}$ are always visible (circumpolar – never visible in south for same lat)
- Sources with $\text{dec} < \text{lat} - 90$ are never visible (circumpolar in south for same lat)

For southern hemisphere (lat is absolute value here):

- Sources with $\text{dec} < \text{lat} - 90$ are always visible (circumpolar)
- Sources with $\text{dec} > 90 - \text{lat}$ are never visible

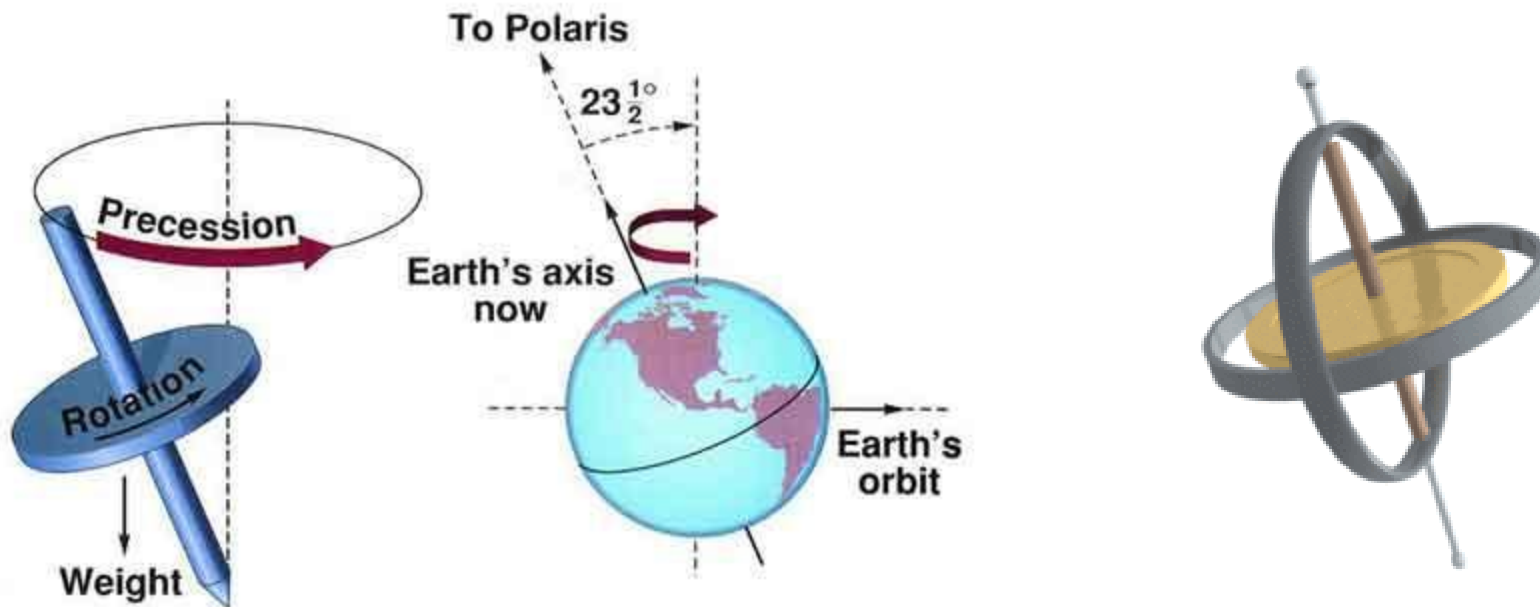
Right Ascension and Hour Angle



Hour Angle (HA) is the angle between the celestial meridian (from horizontal system) to the object's RA

- If an object is transiting, the HA is zero
- Objects before transit have negative HA
- Objects after transit have positive HA
- As time passes, the HA increases
- When will an object with $HA = -2h$ transit?

Right Ascension and Precession



The Equatorial system needs an “Equinox,” which specifies the location of the pole (and vernal equinox) for a given year.

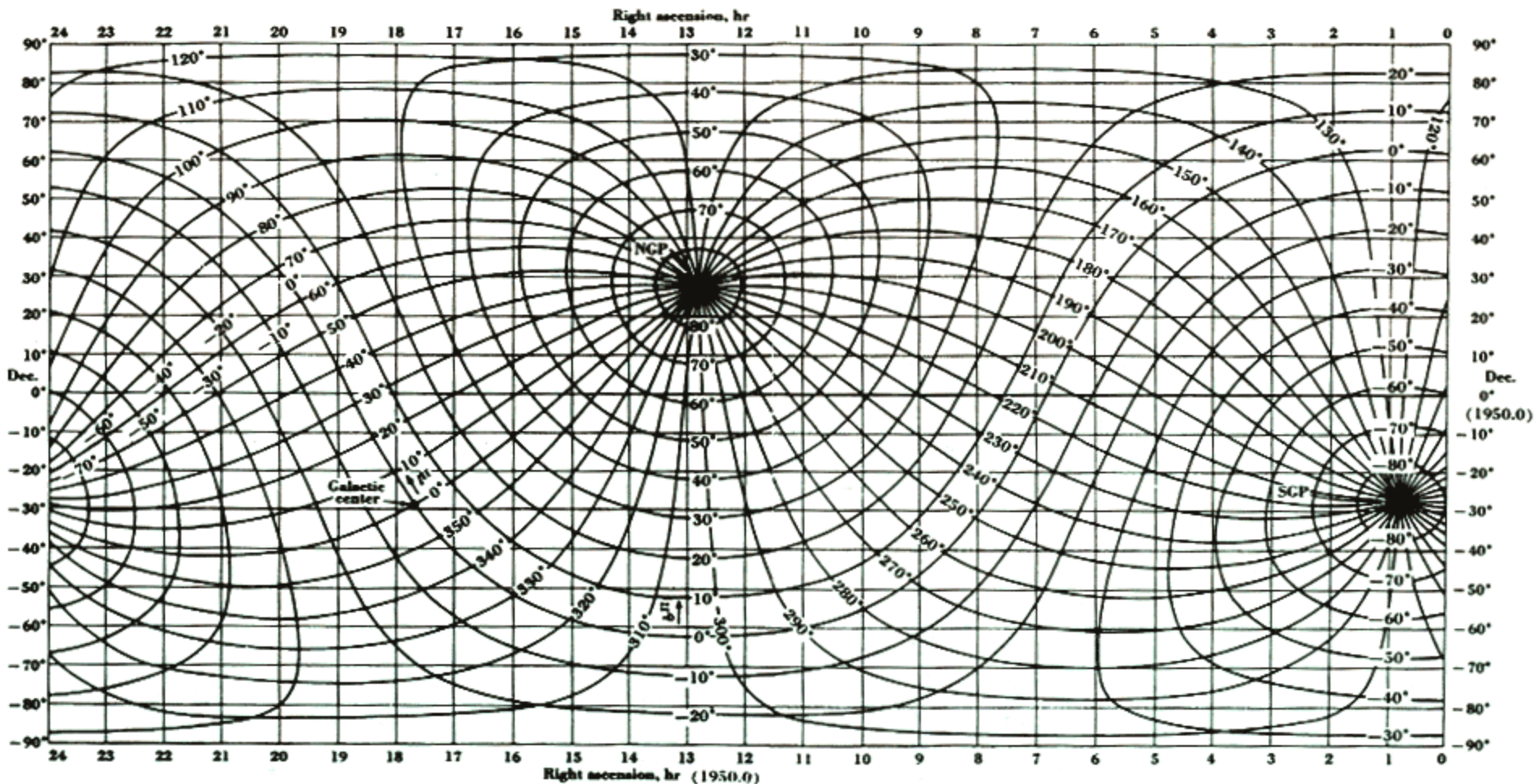
We are currently using “J2000,” but “B1950” is still common, and “J2050” will become popular around 2025 I guess.

The reason why we need this complication will not be completely clear until we talk about time

Converting Between Coordinate Systems

Equatorial and Galactic

Use coordinate converters for accuracy of course

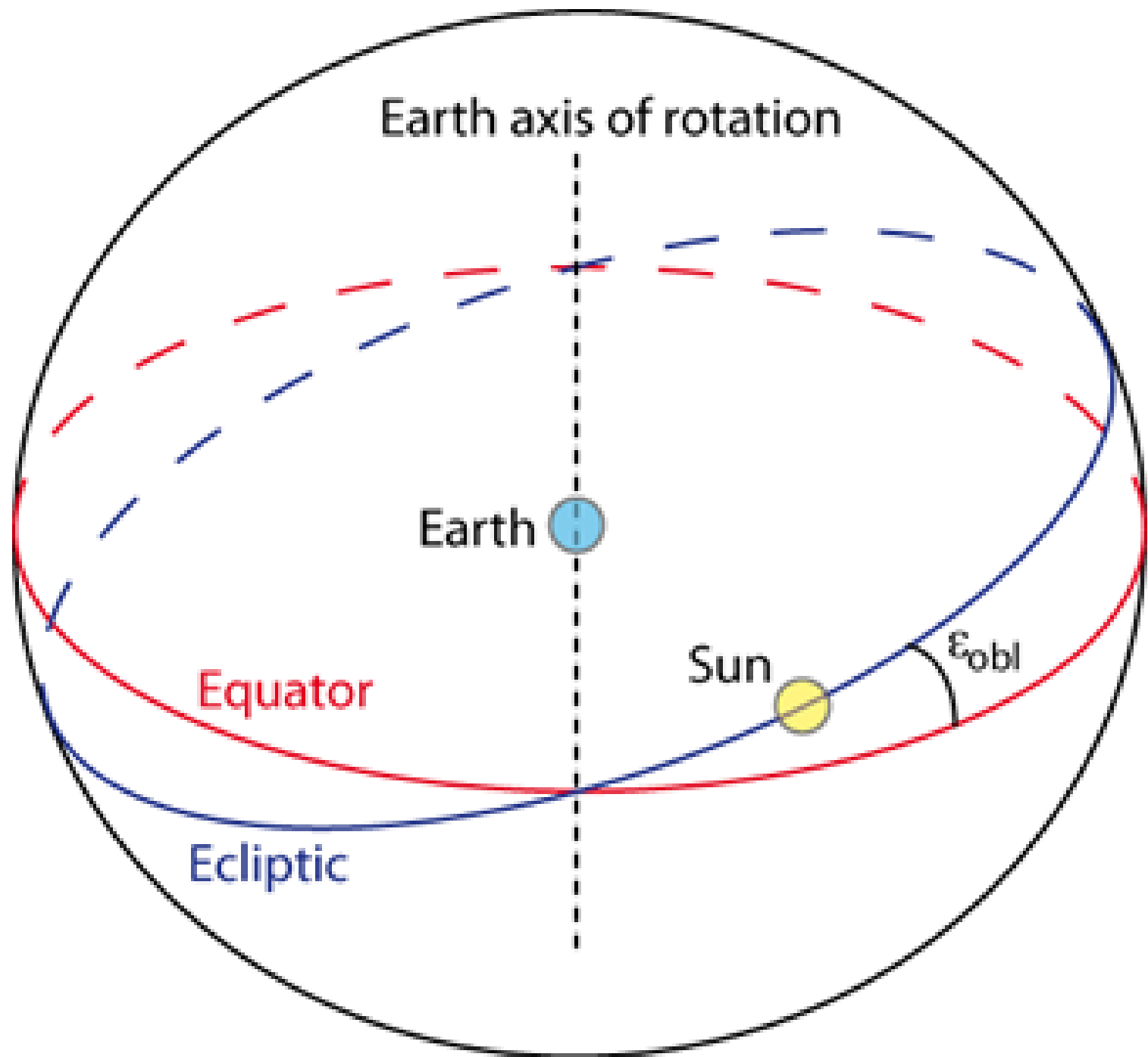


Converting Between Coordinate Systems

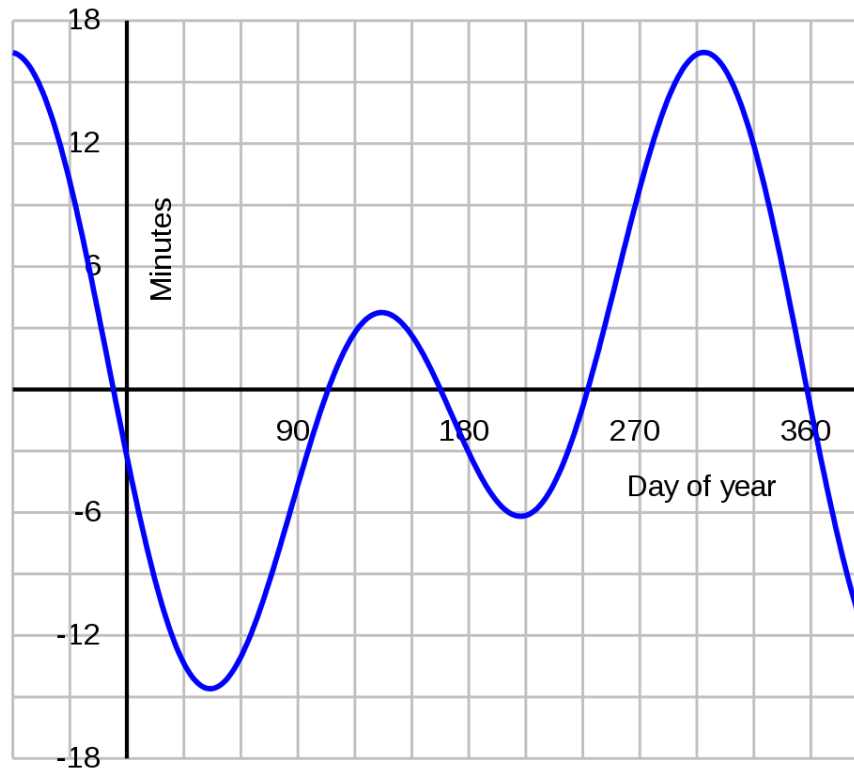
Equatorial and Horizontal – gives rise/set times!

$$\begin{aligned}\cos(a) \sin(A) &= -\sin(HA) \cos(\delta) \\ \sin(a) &= \sin(\delta) \sin(\varphi) + \cos(\delta) \cos(HA) \cos(\varphi)\end{aligned}$$

Here φ is the observer's latitude, HA is hour angle, a is altitude, and A is azimuth.

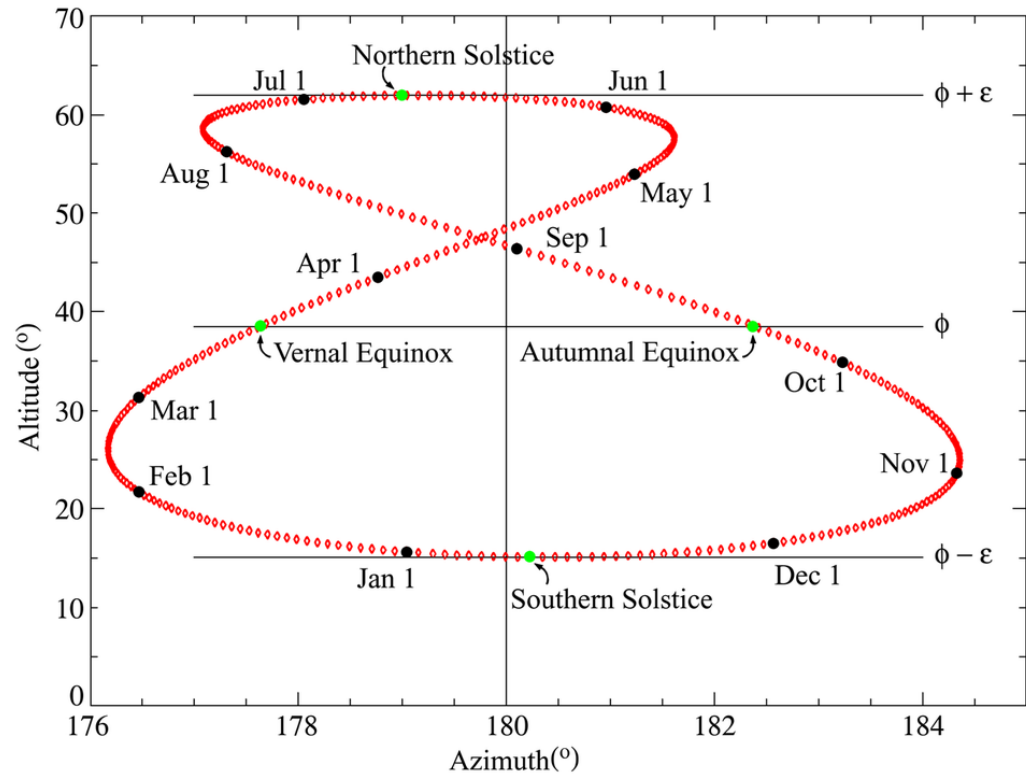
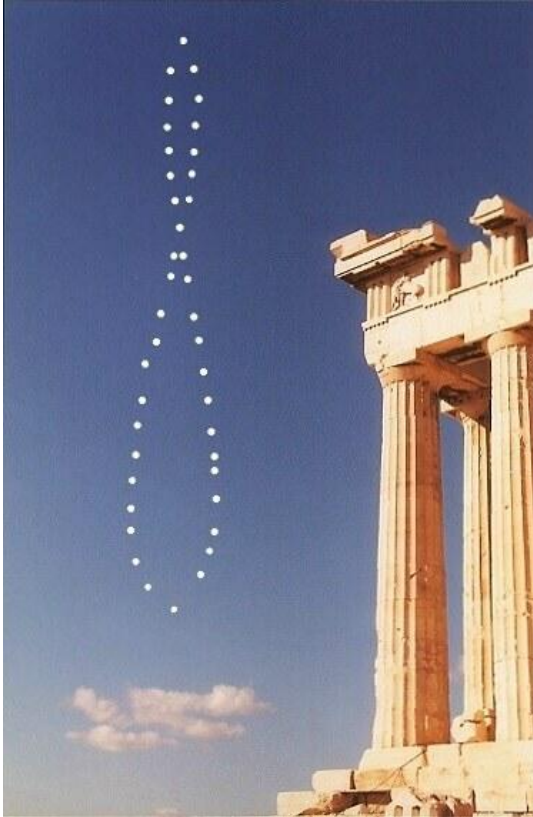


Equation of Time



Tells you the amount of time the Sun is running fast or slow (the annalema) throughout the year.

Annalemma



The figure you get when you take a picture every day at the same time throughout the year.

Sometimes the Sun runs “fast,” sometimes “slow”

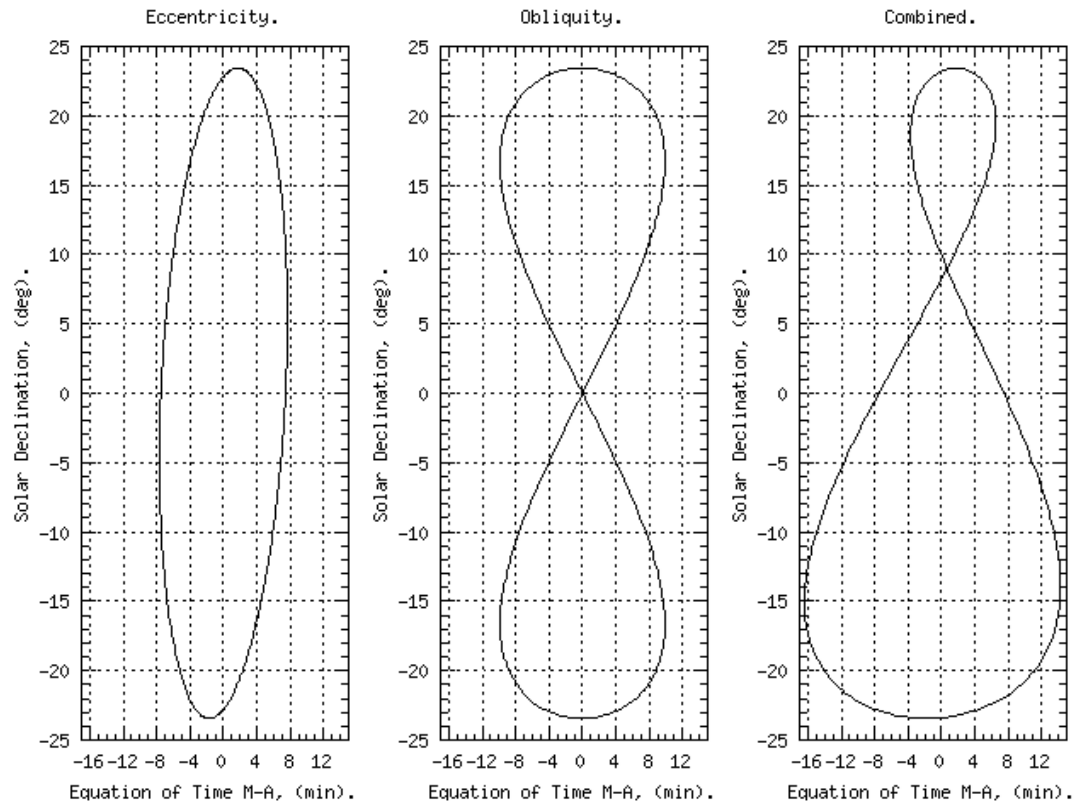
Annalema

Shape caused by:

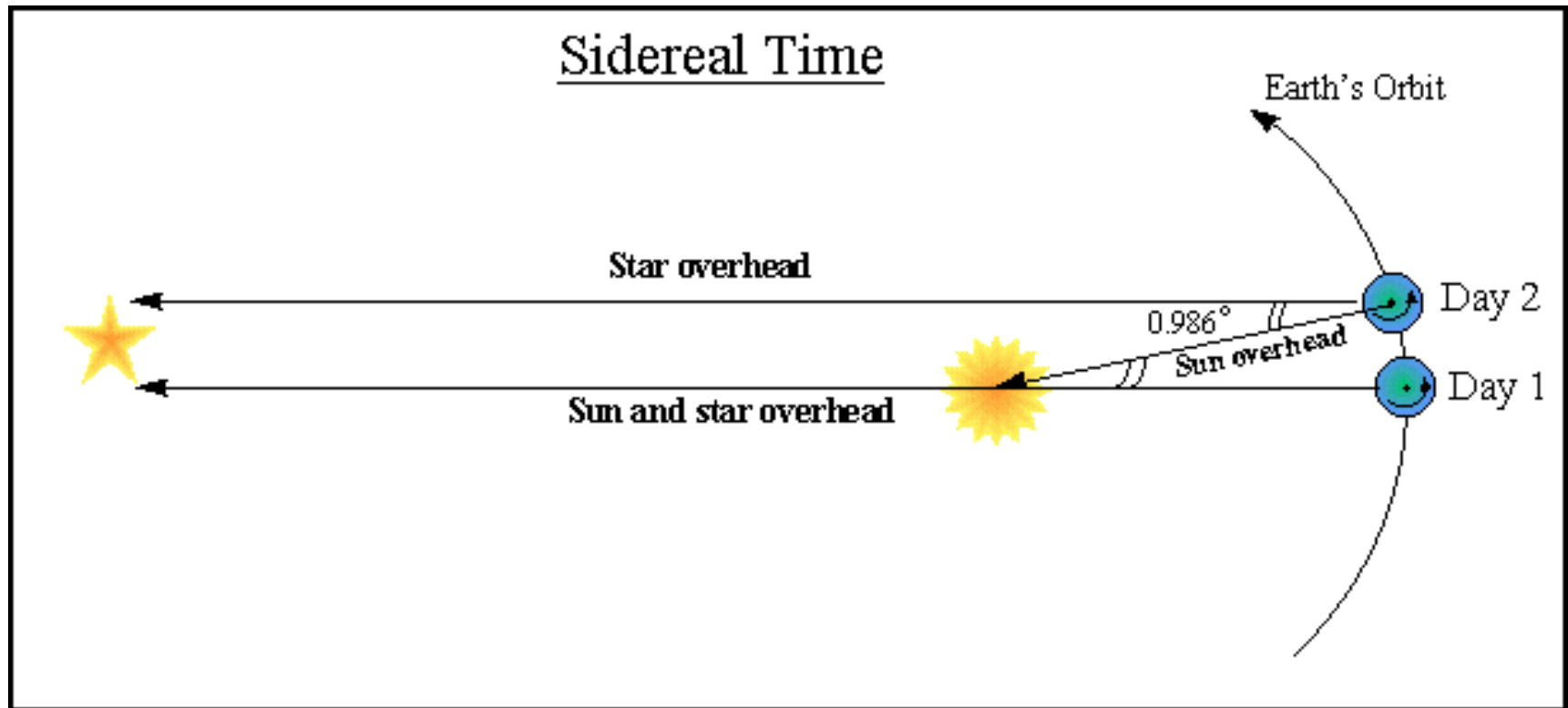
Eccentricity of Earth's orbit

Tilt of Earth's axis ("obliquity")

Tilt of annalema due to fact that Earth's closest, furthest distance from Sun not exactly at the equinoxes.



Sidereal Time



A sidereal day is about 23h 56m long.

Every sidereal day the Sun is at the same place in the sky.

The Sun is found 4 minutes of RA different each day. 2h per month = 4m per day.

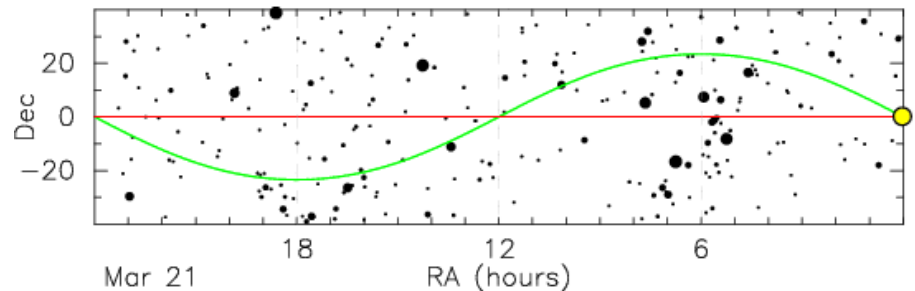
Local Sidereal Time (LST)

The RA of an object on your meridian is by definition the LST!!

$$HA = LST - RA$$

LST is the hour angle of the vernal equinox at your location

The Sun is found 2 hours advanced in RA each month, SO THIS IS THE DIFFERENCE BETWEEN LST and “normal” time!



LST Calculators

<https://aa.usno.navy.mil/data/siderealtime>

Enter the date, UT1 time, and longitude.

Easiest to remove the longitude complication, for comprehension.

On the Vernal Equinox, at Greenwich England, the Sun is at 0h RA. The RA of an object on your meridian is the LST, so when the Sun is on your meridian (noon), the LST is 0h.

LST must catch up 24 hours a year, so 6 months later on the fall equinox the LST and Solar time must be identical.

Adding longitude complicates things further!